



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ANNAPURNA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

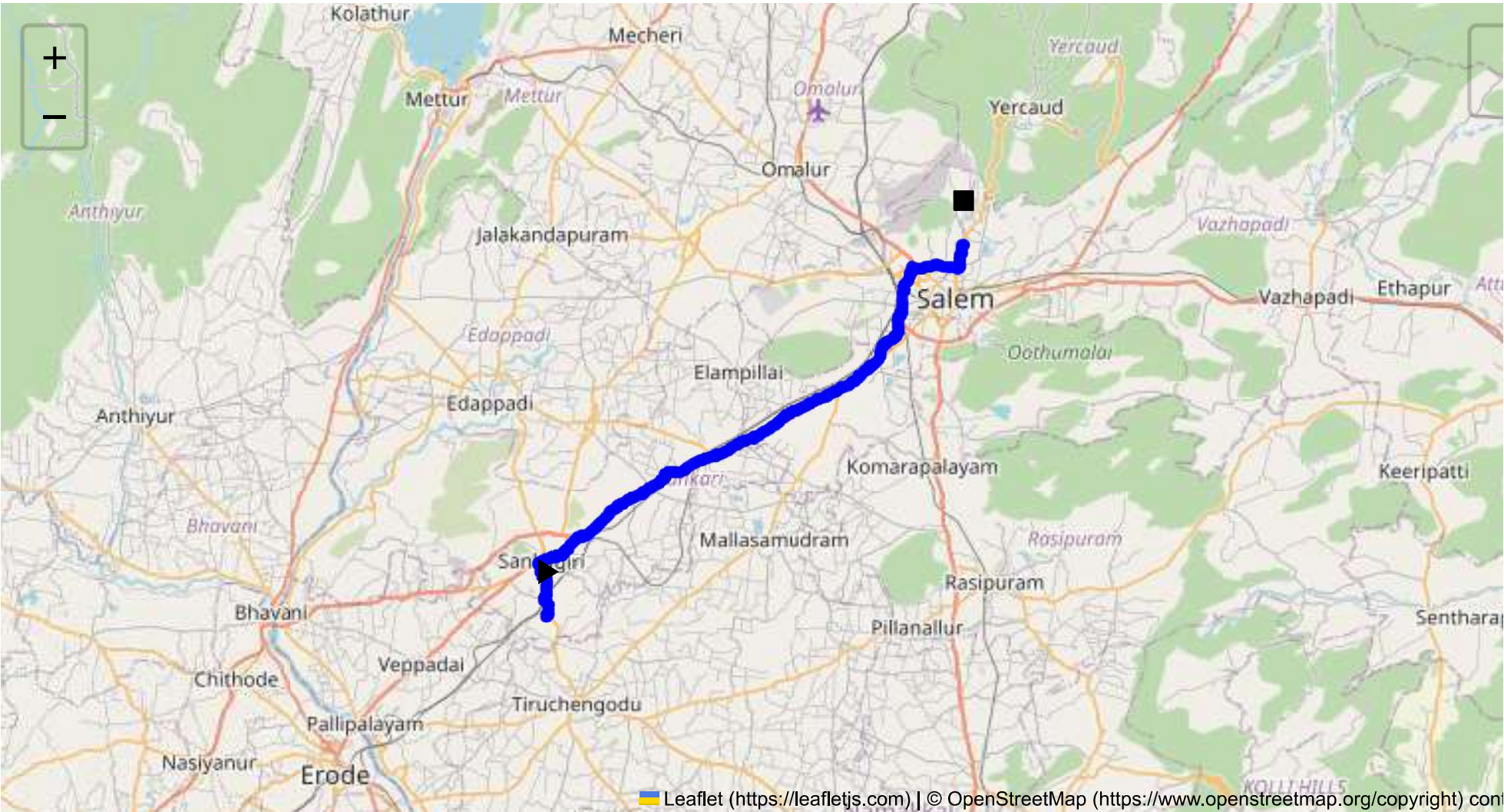
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.39 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.691435, 78.162666)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Chinna Thirupathi, Salem, covers approximately 48.39 kilometers. It typically involves traveling through rural areas, small towns, and parts of urban Salem. The primary roads include state highways and local roads that connect Sangagiri to the main city of Salem.

2. Typical Weather Conditions and Hazards

The weather in this region is generally hot and dry, with a high likelihood of monsoon rains between June and September. During monsoon season, roads can become slippery and may face temporary waterlogging. Dust storms and dry conditions could also affect visibility in summer months.

3. Analysis of Traffic Patterns

Traffic is typically moderate, but it can become congested, particularly around urban areas near Salem. Peak traffic hours are generally between 8:00-10:00 AM and 5:00-7:00 PM. Rural sections may see less congestion but may have unpredictable livestock or slow-moving vehicles.

4. Assessment of Road Quality and Infrastructure

Road quality varies, with some well-maintained stretches and others with potholes and narrow lanes. Infrastructure such as signage, lighting, and guardrails may be inadequate in rural areas. Overbridge or bypass alternatives may be missing, especially in smaller towns.

5. Suggestions for Alternative Routes

In case of emergencies or blockages, one can consider alternate routes through Kollapatty or via NH44, although these might increase travel time. It is advisable to have updated GPS and local traffic alerts for better route planning.

6. Local Regulations for Hazardous Materials

Transporting hazardous materials requires adherence to Indian government regulations, including securing all necessary permits. Vehicles must display appropriate hazard symbols, and drivers need to be trained and licensed for hazardous cargo.

7. Historical Incidents

There have been occasional incidents involving heavy vehicles, often related to poor road conditions or inclement weather. Precaution is required in areas with steep gradients or sharp turns.

8. Environmental Considerations and Sensitive Areas

There are ecological sensitive zones near forested areas closer to Yercaud. Special care should be taken to prevent any spillages that might affect local flora and fauna. Avoid unnecessary honking in wildlife zones.

9. Communication Coverage

Most of the route has good mobile coverage, though rural sections may experience temporary signal drops. It is advisable for drivers to carry a mobile phone from multiple network providers as a backup.

10. Estimated Emergency Response Times

Response times can be quick in Salem with proximity to major hospitals and fire stations. However, rural sections may face longer response times of up to 30-45 minutes due to challenging terrain and limited infrastructure.

11. Overall Summary of Risk Assessment

The route presents moderate risks primarily due to variable road conditions, weather hazards, and traffic congestion in urban centers. Proper planning, adherence to regulations, and real-time monitoring of weather and traffic conditions can mitigate these risks. Emergency backups and awareness of local conditions are essential for safe travel.

This comprehensive risk assessment requires constant updates as conditions on the ground can change rapidly due to weather, construction, and policy updates. Regular engagements with local authorities and transport departments are recommended.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.67557, 78.15976	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.45004, 77.87206	15 KM/Hr
6	Turn	Medium	11.45028, 77.87202	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47410, 77.86845	30 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.60942, 78.10056	30 KM/Hr
12	Turn	Medium	11.61736, 78.10588	30 KM/Hr
13	Turn	High	11.64007, 78.11794	15 KM/Hr
14	Turn	Medium	11.64011, 78.11789	30 KM/Hr
15	Turn	High	11.67074, 78.12668	15 KM/Hr
16	Turn	High	11.67081, 78.12661	15 KM/Hr
17	Turn	High	11.67582, 78.12786	15 KM/Hr
18	Blind Spot	Blind Spot	11.69136, 78.16293	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
26	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
27	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
29	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
30	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
31	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
32	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
33	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
34	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
35	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
36	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
37	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
38	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
39	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
40	clinic	Ayush Multispeciality Homoeopathic Clinic	11.6770982, 78.1395947	30 km/h	Medium
41	police	Alagapuram Police station	11.6769953, 78.1422615	30 km/h	Medium
42	hospital	AP medical center	11.6774074, 78.1425988	30 km/h	Medium
43	hospital	M G Diabetes Speciality and Research Centre	11.6766146, 78.1422986	30 km/h	Medium
44	hospital	Nathan Super Speciality Hospital	11.6743874, 78.1417222	30 km/h	Medium
45	hospital	Devi Hospital	11.6786391, 78.1415731	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
46	hospital	Fairlands Eye Cure	11.677959, 78.1415897	30 km/h	Medium
48	hospital	Dr. Soodamani Hospital	11.6757031, 78.14473	30 km/h	Medium
49	hospital	Kavin Eniyan Hospital	11.677643, 78.144129	30 km/h	Medium
50	hospital	Sri Vaari Hospital	11.6763726, 78.1467277	30 km/h	Medium
51	hospital	Krishna Speciality Hospital	11.676751, 78.145561	30 km/h	Medium
54	hospital	Gopi Memorial Hospital	11.673855, 78.1516078	30 km/h	Medium
55	hospital	Salem Gopi Hospitals	11.6745894, 78.1515482	30 km/h	Medium
56	hospital	Dr. P. Sundarrajan Neuro Hospital	11.6742791, 78.1499738	30 km/h	Medium
57	hospital	Mahadevas Ayurvedic Hospital	11.6752045, 78.1517396	30 km/h	Medium
58	clinic	Akshaya Fertility Centre	11.6739088, 78.1518657	30 km/h	Medium
59	clinic	Saraswathi Nursing Home	11.6746327, 78.1520981	30 km/h	Medium
60	hospital	J S Hospital, Dharmapuri	11.6739064, 78.1521557	30 km/h	Medium
62	hospital	Shanmuga Hospital and Salem Cancer Institute	11.6770178, 78.1531844	30 km/h	Medium
63	hospital	Tms Eye Hospital	11.6772791, 78.1536202	30 km/h	Medium
64	hospital	Sureka Hospital	11.676423, 78.1550511	30 km/h	Medium
66	hospital	Ramana Surgical Hospital	11.6731205, 78.15862	30 km/h	Medium
67	hospital	Salem Medical Centre	11.67607, 78.163014	30 km/h	Medium
68	hospital	KS Siddha Hospital	11.6752685, 78.1633063	30 km/h	Medium
69	clinic	Sindhu Eye Care Centre	11.6814684, 78.1625895	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
28	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
47	school	Saradha Balamandir School	11.6771182, 78.1472461	30 km/h	Medium
52	school	Saradha College	11.6769525, 78.1481901	30 km/h	Medium
53	school	Sri Sarada Matriculation Higher Secondary School	11.6768969, 78.1488115	30 km/h	Medium
61	school	Holy flower matric hr.sec school	11.6748637, 78.1528099	30 km/h	Medium
65	school	CSI Hobart Edu Institute	11.673245, 78.1567934	30 km/h	Medium
70	college	Govt ITI	11.6919383, 78.1636731	30 km/h	Medium

Route Photos of Risky Spots